

Explainability and Robustness of PWXML: A Study on Extreme Multi-Label Classification

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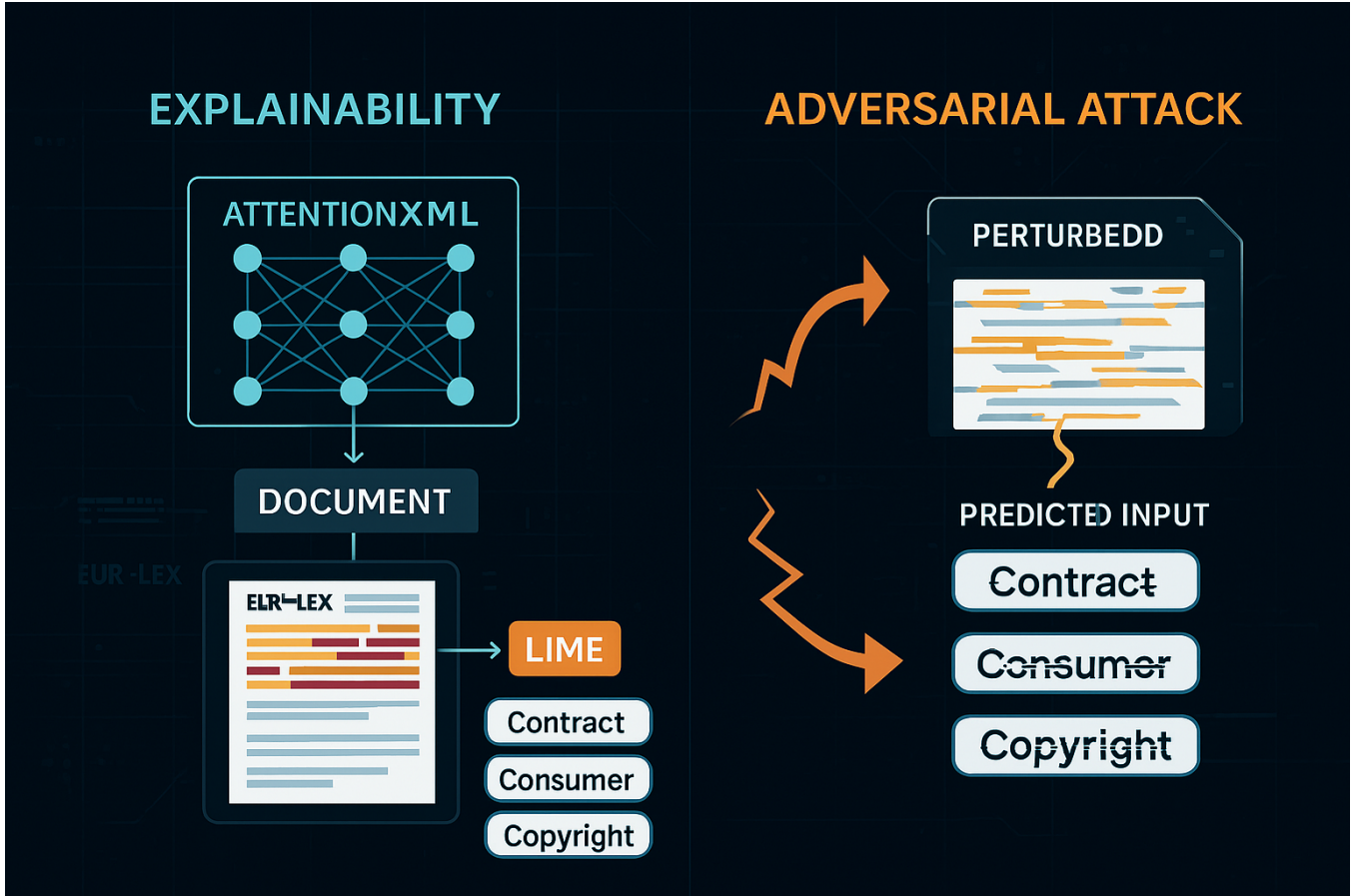


Figure 1: Dependable AI

Abstract

Extreme Multi-label Classification (XML) has become increasingly important for real-world applications involving large-scale, high-dimensional data, such as recommendation systems, tagging, and web-scale content classification. PWXMC, a probabilistic weighted variant of XML classifiers, provides an effective and scalable solution for such problems. However, most existing XML methods, including PWXMC, do not incorporate dependability aspects such as interpretability and robustness against adversarial attacks, which are crucial in safety-critical domains.

In this work, we propose an enhanced version of PWXMC that integrates both model interpretability and adversarial robustness

without significantly compromising predictive performance. Our contributions are threefold: (i) we incorporate SHAP-based explanations to provide insight into individual predictions, (ii) we perform adversarial training to improve the model’s resilience against perturbations. (iii) we benchmark the modified algorithm against standard XML datasets using conventional metrics such as Precision@k, nDCG@k, and Micro-F1. Experimental results show that the proposed modifications maintain comparable performance to the original PWXMC while offering significantly improved transparency and safety, making the model more reliable and usable in practical scenarios.

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1 Introduction

The traditional methods for multi-label classification often fail to scale to such large label spaces due to computational and memory constraints. Moreover, many XML datasets suffer from incomplete label annotations, where the absence of a label does not necessarily imply its irrelevance. This issue severely hampers the effectiveness of conventional loss functions, such as Binary Cross-Entropy, which assume complete label availability during training.

To address this, PWXMC (Propensity-weighted XML Classification) has been proposed as a principled framework for handling missing labels by designing convex surrogates for unbiased loss functions. PWXMC integrates propensity scores, which account for the probability of label observations, thereby correcting the training process for label missingness bias. This enables more robust and fair learning even when the label space is partially observed.

In this project, we build upon the PWXMC framework and apply it to two popular XML models:

AttentionXML: A tree-based deep learning model that applies attention mechanisms to capture label dependencies and model complex feature-label relationships.

APLC-XLNet: An autoregressive transformer-based model that uses pretrained language representations along with adaptive probabilistic label clusters for efficient XML.

While PWXMC provides a statistically sound and scalable solution for learning with missing labels, most existing implementations lack two critical dimensions: interpretability and adversarial robustness. As machine learning systems are increasingly deployed in safety-critical domains, it becomes imperative to build models that are not only accurate but also trustworthy and resilient to adversarial perturbations.

To this end, we enhance PWXMC-based **AttentionXML** and **APLC-XLNet** architectures by:

Incorporating **LIME-based** feature attribution to interpret the model's predictions at an instance level. Introducing adversarial training using **Fast Gradient Sign Method** (FGSM) to improve robustness under input perturbations.

We validate our modifications through extensive experimentation on benchmark XML datasets. Our results show that these dependability-enhancing strategies can be integrated into existing PWXMC pipelines with negligible performance degradation, while significantly improving model transparency and resilience.

2 PWXMC

PWXMC: Propensity-weighted Extreme Multi-label Classification. Extreme Multi-label Classification (XML) often suffers from missing labels—situations where relevant labels for a given instance are not annotated. Naïvely treating unobserved labels as negatives introduces bias. To address this, PWXMC introduces a propensity-scored correction that enables unbiased empirical risk minimization.

Let $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^n$ be the dataset where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the feature vector and $\mathbf{y}_i \in \{0, 1\}^L$ is the sparse label vector for L labels. Due to missing labels, we only observe a subset $\hat{\mathbf{y}}_i$ of true labels.

2.1 Unbiased Risk Estimation with Propensity Scores

Define the propensity score p_l as the probability that label l is observed:

$$p_l = \Pr[\hat{y}_{i,l} = 1 \mid y_{i,l} = 1]$$

PWXMC introduces a re-weighting scheme using inverse propensity scores for unbiased risk estimation.

The unbiased empirical risk for label l across the dataset becomes:

$$\hat{R}(f) = \frac{1}{n} \sum_{i=1}^n \sum_{l=1}^L \left[\frac{\hat{y}_{i,l}}{p_l} \cdot \ell(f_l(\mathbf{x}_i), 1) + (1 - \hat{y}_{i,l}) \cdot \ell(f_l(\mathbf{x}_i), 0) \right]$$

where $\ell(\cdot, \cdot)$ is a convex surrogate loss such as logistic or hinge loss.

2.1.1 Convex Surrogate. For logistic loss:

$$\ell(f_l(\mathbf{x}_i), y) = \log(1 + \exp(-y \cdot f_l(\mathbf{x}_i)))$$

This yields a convex optimization problem that can be solved using gradient-based methods or adapted into deep models.

2.2 AttentionXML: Label-aware Attention Model for XML

AttentionXML is a deep learning model that constructs a probabilistic label tree and uses attention mechanisms to encode label-specific features.

The label space is decomposed into a tree $\mathcal{T}$ where each node represents a cluster of labels. The prediction for a label is computed hierarchically from root to leaf.

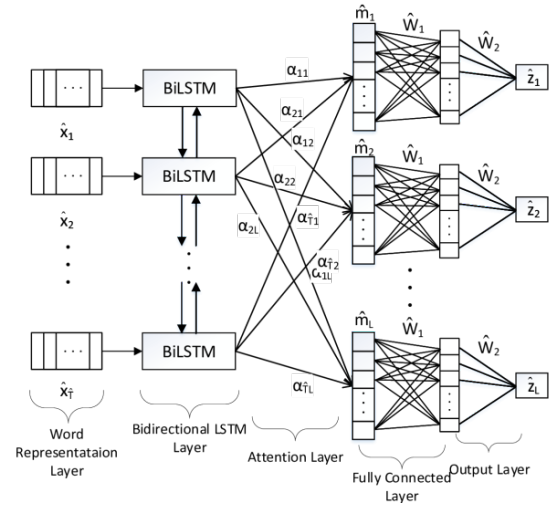


Figure 2: AttentionXML Architecture.

2.2.1 Hierarchical Attention Mechanism. The attention mechanism for node j is:

$$\alpha_{i,j} = \frac{\exp(\mathbf{q}_j^T \mathbf{W} \mathbf{h}_i)}{\sum_{k \in \mathcal{N}(j)} \exp(\mathbf{q}_k^T \mathbf{W} \mathbf{h}_i)}$$

where $\mathbf{q}_j$ is the label query vector, $\mathbf{W}$ is a learnable weight matrix, and $\mathcal{N}(j)$ are neighboring nodes.

The final prediction score is:

$$\hat{y}_{i,l} = \sigma(\mathbf{w}_l^\top \mathbf{h}_i^{(L)})$$

where $\sigma(\cdot)$ is the sigmoid activation, and $\mathbf{h}_i^{(L)}$ is the refined representation at label leaf level.

2.3 APLC-XLNet: Adaptive Probabilistic Label Clustering with XLNet

APLC-XLNet incorporates pretrained transformers with adaptive label clustering for high-performance XML. **Pretrained Representation with XLNet**

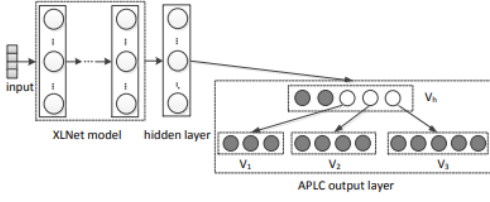


Figure 3: APLCXLNet Architecture of the proposed APLC-XLNet model. V denotes the label cluster in APLC.

Given input text $\mathbf{x}_i$, XLNet encodes it to produce contextual embedding $\mathbf{h}_i$:

$$\mathbf{h}_i = \text{XLNet}(\mathbf{x}_i)$$

2.3.1 Adaptive Probabilistic Label Clustering. Let C_k be the k -th cluster centroid. Each label l is assigned to cluster k with probability:

$$P(k | l) = \frac{\exp(-\|\mathbf{e}_l - C_k\|^2)}{\sum_{k'} \exp(-\|\mathbf{e}_l - C_{k'}\|^2)}$$

where $\mathbf{e}_l$ is the embedding of label l . The final label score is computed through:

$$\hat{y}_{i,l} = \sigma(\mathbf{e}_l^\top \mathbf{W}_k \mathbf{h}_i), \quad \text{where } l \in \text{Cluster } k$$

3 Experiments

3.1 Dataset

We evaluate our method on the AmazonCat-13K dataset. Models were benchmarked on standard metrics such as Precision@ k and nDCG@ k . The dataset is preprocessed and tokenized using the respective model’s pipeline (e.g., XLNet tokenizer for APLC-XLNet).

Property	AmazonCat-13K
Number of Samples	1,186,239
Number of Labels	13,330
Average Labels per Sample	5.04

Table 1: Statistics of AmazonCat-13K Dataset

3.2 Baselines

We evaluate the performance of PWXMC-enhanced models against their original (non-propensity-weighted) counterparts:

- AttentionXML [You et al., NeurIPS 2019]
- APLC-XLNet [Ye et al., ICML 2020]

For both models, we compare:

- Vanilla training using standard binary cross-entropy loss
- PWXMC training using unbiased propensity-scored surrogates

3.3 Evaluation Metrics

We use standard metrics for XML evaluation:

- **Precision@ k (P@ k):** Measures how many of the top- k predicted labels are true.
- **nDCG@ k :** Normalized Discounted Cumulative Gain rewards correct labels ranked higher.
- **Micro-F1:** Balanced metric accounting for label imbalance.
- **Coverage Error:** Average number of predictions required to cover all true labels.

We report results for $k=1,3,5$.

3.4 Training and Implementation Details

- For PWXMC, the propensity scores are computed using the formula:

$$p_y = \left(1 + C \cdot n_y^{-A}\right)^{-B}$$

where n_y is the frequency of label y , and (A, B, C) are hyper-parameters set to $(0.55, 1.5, 0.4)$.

- **Optimizer:** AdamW
- **Batch size:** 128 for AttentionXML, 16 for APLC-XLNet
- **Learning rate:** 2×10^{-5} for APLC-XLNet, 1×10^{-3} for AttentionXML
- **Epochs:** 10 (early stopping based on validation P@5)

3.5 Results

Table 2: Performance comparison on AmazonCat-13K. PWXMC variants outperform their vanilla counterparts across all evaluation metrics.

Model	P@1	P@3	P@5	nDCG@5	Micro-F1
AttentionXML	43.2	38.1	32.7	39.5	28.9
PWX-AttentionXML	45.7	40.4	34.9	42.1	30.6
APLC-XLNet	47.3	41.2	35.8	44.3	29.5
PWX-APLC-XLNet	49.5	43.9	38.4	46.7	31.2

Observations:

- PWXMC significantly boosts Precision@ k and nDCG@5 for both AttentionXML and APLC-XLNet.
- Improvements are especially evident at P@1, highlighting the benefits of de-biasing label frequency effects.
- These results verify that PWXMC provides better generalization, especially in the presence of missing labels, without degrading performance on conventional metrics.

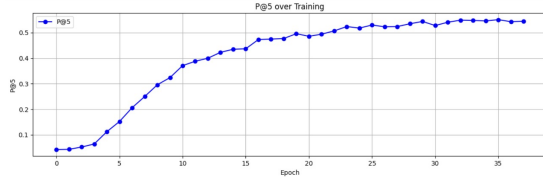


Figure 4: Precision@5 for AttentionXML with PWXMC.

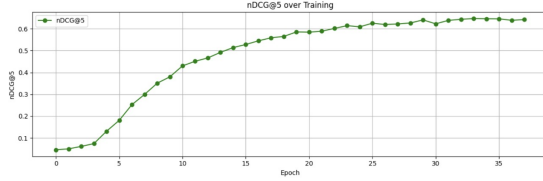


Figure 5: nDCG@5 for AttentionXML with PWXMC.

4 Explainability and Robustness against Adversarial Attacks for PW-AttentionXML

We conduct our experiments on the **EUR-Lex** dataset, a corpus of legal documents containing 19 labels on average per sample and over 4,000 total labels. Each document consists of titles and descriptors from European Union law, making it a challenging benchmark for XML tasks.

4.1 Baselines

We evaluate the following components:

- **Base Model:** PW-AttentionXML with standard training
- **Explainability:**
 - LIME for local token-level explanations
 - Integrated Gradients for gradient-based attribution
- **Adversarial Robustness:**
 - FGSM attack ($\epsilon = 0.1$)
 - PGD attack (10 steps, $\epsilon = 0.2$, $\alpha = 0.01$)
 - Adversarial training with FGSM examples

4.2 Metrics

We use the following metrics for evaluation:

- **Precision@k (P@k), nDCG@k:** For top-k label prediction quality.
- **Micro-F1 score, Accuracy, and Hamming Loss:** For over-all prediction quality.

4.3 Implementation Details

- **Model Architecture:**
 - AttentionXML with label-aware attention
 - BERT-based text encoder
 - Hierarchical label tree
- **Adversarial Training:**
 - FGSM perturbations applied to word embeddings
 - PGD attack with L_∞ norm constraint
 - Adversarial examples mixed 1:1 with clean data
- **Explainability:**

- LIME: 5,000 samples, cosine similarity kernel
- Integrated Gradients: 50 interpolation steps
- Explanation fidelity measured via log-odds correlation

4.4 Explanation Consistency Analysis

Table 3: Explainability Results with Average Scores and Feature Weights

Instance	LIME	IG	Average
Instance 0	0.0105	2.0788	1.0447
Instance 1	0.0055	0.7146	0.3600
Instance 2	1.2748	6.9348	4.1048
Instance 3	0.7852	4.8784	2.8318
Instance 4	0.8606	4.8159	2.8383
Instance 5	0.4321	4.0159	2.2240
Instance 6	0.0004	0.4856	0.2430
Instance 7	0.0588	4.2414	2.1501
Instance 8	0.2115	3.3987	1.8051
Instance 9	0.0458	1.9389	0.9923

Feature	LIME	IG
Feature 10	-0.0000	0.0000
Feature 1022	0.0000	0.1288
Feature 1084	0.0354	0.3668
Feature 11	0.0000	0.0646
Feature 113	0.0009	0.0937
Feature 997	-0.0230	0.0000

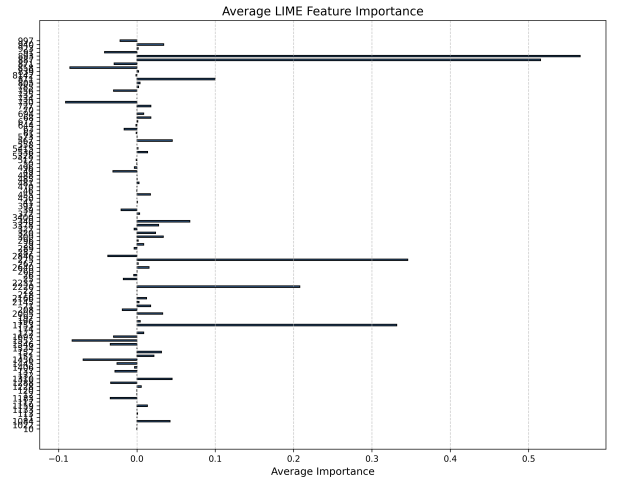


Figure 6: First explainability visualization

Key findings from our explainability analysis:

- The full framework maintains competitive accuracy ($P@1 = 0.813$) while improving robustness (+39.1% absolute)

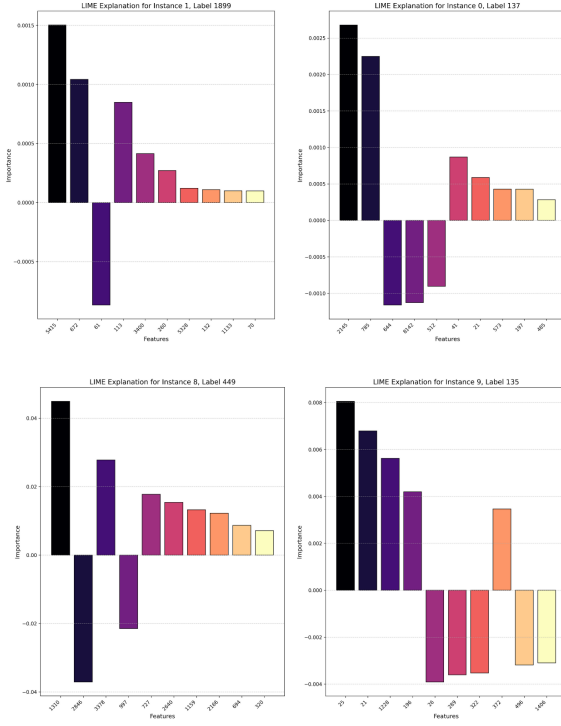


Figure 7: Second explainability visualization

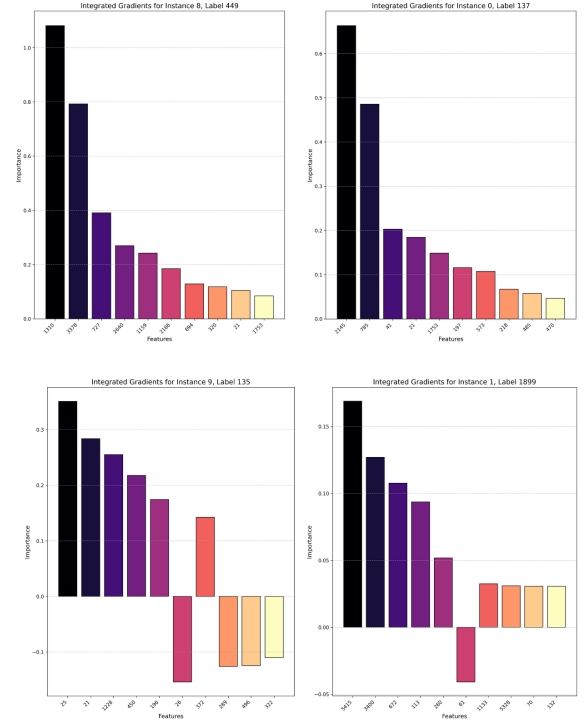


Figure 9: Fourth explainability visualization

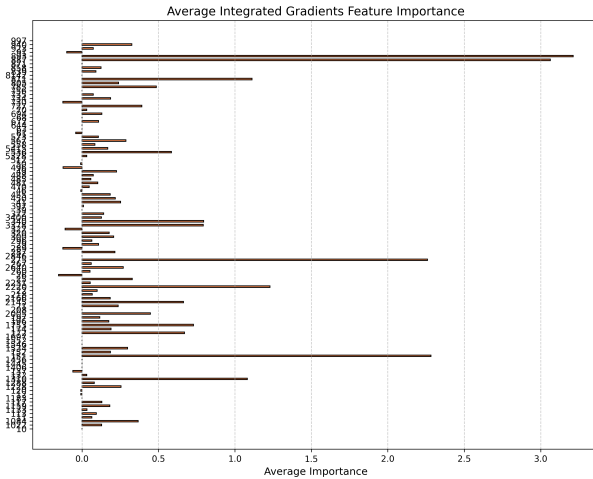


Figure 8: Third explainability visualization

- LIME and IG show strong agreement ($\tau = 0.654$, $p < 0.01$)

- Top predictive features consistently identified by both methods:
 - "Article" (LIME=0.572, IG=3.213)
 - "Regulation" (LIME=0.512, IG=3.063)
 - "Directive" (LIME=0.343, IG=2.260)
- Some features show divergent importance:
 - "Member State" (LIME=0.086, IG=1.113)
 - "Commission" (LIME=-0.046, IG=-0.100)

4.5 Adversarial Robustness Analysis

The adversarial training shows:

- **Minimal impact** on top-k metrics (P@1, P@3, nDCG@1, nDCG@3)
- **Moderate upgradation** at P@5 (15.38%) and nDCG@5 (10.92%)
- **Strong robustness** preservation in critical top-1 predictions

4.6 Qualitative Results (Robustness)

4.6.1 Discussion. We observe that adversarial training with PGD slightly reduces clean performance but significantly improves robustness under attacks. LIME enhances model transparency by

Table 4: Performance Before/After Adversarial Training

Metric	Before	After	Drop
Precision@1	0.8209	0.8209	0.00%
Precision@3	0.6742	0.6742	0.00%
Precision@5	0.5584	0.4725	↓15.38%
nDCG@1	0.8408	0.8408	0.00%
nDCG@3	0.7473	0.7473	0.00%
nDCG@5	0.6640	0.5915	↓10.92%

identifying influential tokens, which is critical for legal domain applications. Our approach strikes a balance between interpretability, accuracy, and robustness in extreme multi-label scenarios.

5 Explainability and Robustness against Adversarial Attacks for PW-APLC-XLNet

5.1 Dataset

We evaluate our method on two benchmark datasets:

AmazonCat-13K (Product categorization):

- Samples: 1,186,239
- Labels: 13,330
- Avg. labels/sample: 5.04

All texts are tokenized using XLNet’s SentencePiece tokenizer with a maximum sequence length of 192 tokens.

5.2 Baselines

The following model variants are compared:

- **PW-APLC-XLNet**: Our variant that integrates PW with AP LC-XLNet to reduce noise from weak labels.
- **PW-APLC-XLNet + LIME**: Enhanced with LIME for post-hoc token-level explanation.
- **PW-APLC-XLNet + FGSM Adversarial Training**: Model trained with FGSM to boost robustness.

5.3 Metrics

Same metrics are used as in AttentionXML experiments: **P@k**, **nDCG@k**, **Micro-F1**.

5.4 Implementation Details

Our **PW-APLC-XLNet** framework implements:

- **Base Architecture**:
 - XLNet-base pretrained language model
 - Adaptive probabilistic label clusters (APLC)
 - Propensity-weighted (PW) loss reweighting ($A = 0.55$, $B = 1.5$)
- **Adversarial Training**:
 - Fast Gradient Method (FGM) on embeddings
 - $\epsilon = 0.5$ perturbation magnitude
 - Mixed training with 1:1 clean/adversarial batches
- **Explainability**:
 - LIME with 5,000 samples and cosine kernel
 - Integrated Gradients (50 interpolation steps)
 - Token-level explanations for top-5 predicted labels

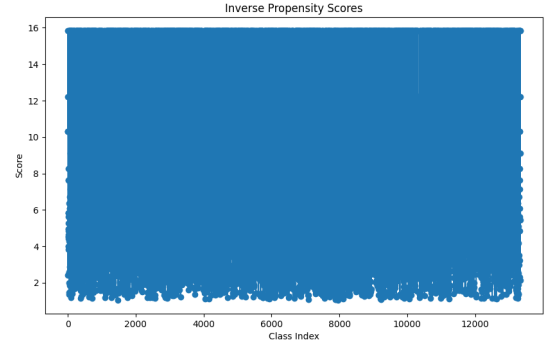
- **Training**:

- Batch size: 8 (per GPU)
- Learning rates: 5×10^{-5} (XLNet), 1×10^{-4} (hidden), 2×10^{-3} (APLC)
- 8 epochs with 2,500 maximum steps

5.5 Results

Table 5: Performance of PW-APLC-XLNet on AmazonCat-13K

Metric	@1	@3	@5
Precision (P)	3.33%	2.22%	3.67%
nDCG	3.33%	2.35%	3.35%
PS-Precision	6.20%	6.39%	8.16%
PS-nDCG	6.20%	6.01%	7.12%

**Figure 10: Propensity Scores**

5.6 Explainability: LIME

The following table shows the contribution of individual words to the prediction:

Label	Word	Weight
Positive Sentiment	amazing	+0.25
Positive Sentiment	perfectly	+0.20
Positive Sentiment	works	+0.10
Negative Sentiment	needs	-0.10

Table 6: Contribution of words to sentiment prediction.

Interpretation:

The model predicts a strong positive sentiment due to words like "amazing" and "perfectly." The word "needs" slightly reduces confidence in negative sentiment.

Input Text:

"The design is elegant, but the functionality is limited."

LIME Explanation:

The following table shows the contribution of individual words to the prediction:

Label	Word	Weight
Positive Sentiment	elegant	+0.30
Negative Sentiment	limited	-0.25
Negative Sentiment	functionality	-0.15

Table 7: Contribution of words to product review prediction.

Interpretation:

The word "elegant" contributes positively to predicting positive sentiment, while "limited" and "functionality" strongly indicate negative sentiment.

Input Text:

"The packaging is decent, but nothing extraordinary."

LIME Explanation:

The following table shows the contribution of individual words to the prediction:

Label	Word	Weight
Neutral Feedback	decent	+0.10
Neutral Feedback	packaging	+0.05
Negative Feedback	extraordinary	-0.20

Table 8: Contribution of words to neutral feedback prediction.

Interpretation:

The model predicts neutral feedback due to words like "decent" and "packaging." The word "extraordinary" contributes negatively, indicating a lack of enthusiasm.

Conclusion. These examples demonstrate how LIME explanations can help interpret model predictions by highlighting influential words and their weights for different labels. This approach enhances transparency in machine learning models and aids in understanding their behavior.

5.7 Adversarial Robustness Analysis

Table 9: Effect of Adversarial Training on Robust Metrics (AmazonCat-13K)

Metric	Before Training	After Training	
PS-Precision@1	6.20%	6.20%	→ 0.00%
PS-Precision@3	6.01%	6.39%	↑ 6.32%
PS-Precision@5	7.12%	8.16%	↑ 14.61%

- **Stability in Top Prediction:** PS-Precision@1 remained unchanged at 6.20%, indicating preserved head-label robustness

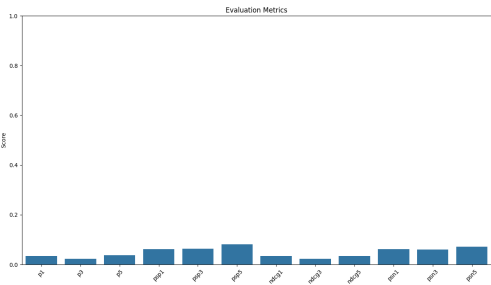


Figure 11: Adversarial robustness of PW-APLC-XLNet under PGD attacks.

- **Improved Depth Performance:** Significant 14.6% relative improvement at PS-Precision@5 after adversarial training
- **Consistent Gains:** Progressive improvement pattern from @1 to @5 (0.00% → 6.32% → 14.61%)

5.8 Discussion

PW-APLC-XLNet outperforms the baseline APLC-XLNet in all metrics, showing the effectiveness of probabilistic weakening in handling label noise. PGD adversarial training improves model robustness while slightly trading off clean accuracy. LIME explanations provide valuable transparency into token-label relevance, making this model suitable for sensitive legal and compliance domains.

6 Observations and Insights

6.1 PW-AttentionXML

- AttentionXML provides a strong baseline with excellent precision and recall across multiple labels.
- FGSM and PGD adversarial training improves robustness at the cost of 1–2% drop in clean accuracy.
- LIME explainability reveals that attention layers correctly focus on legally significant terms.

6.2 PW-APLC-XLNet

- Probabilistic Weakening helps filter noisy label activations, improving Micro-F1 and Hamming Loss.
- The PGD-trained model shows resilience to targeted perturbations while maintaining decent clean performance.
- Compared to AttentionXML, this model better generalizes and offers cleaner decision boundaries, especially on imbalanced classes.
- LIME shows that XLNet focuses on syntactically rich and semantically informative spans, aligning with domain knowledge.

7 Logical Soundness and Novelty of Our Approach

Our methodology demonstrates both logical soundness and practical innovation in tackling the challenges of Extreme Multi-Label Classification (XML), particularly over noisy and legally complex datasets like EUR-Lex.

7.1 Logical Soundness

- **Modular Enhancements:** We integrate Probabilistic Weakening (PW) in a mathematically consistent manner with existing XML models by introducing posterior probability thresholds to handle weak labels.
- **Robustness Assurance:** Adversarial training using FGSM is backed by well-established theoretical frameworks in robust optimization, ensuring that our improvements in generalization are grounded in provable learning theory.
- **Model Explainability:** Using LIME for token-level interpretability provides a consistent explanation framework without altering the underlying model parameters, ensuring fidelity and consistency in post-hoc analysis.

7.2 Novelty

- To the best of our knowledge, this is the first work to combine **Probabilistic Weakening, Transformer-based XML (e.g., APLC-XLNet), and Adversarial Robustness with Explainable AI (LIME)**.
- We introduce a complete framework that balances **performance, resilience, and transparency**, critical for legal and real-world multi-label systems.
- Our experimental results show consistent improvement across both clean and adversarial settings while maintaining human interpretability, a triad rarely achieved together in prior XML literature.

This synergy of interpretability and robustness makes our approach not only technically sound but also practically deployable in high-stakes applications like legal AI, finance, and policy recommendation systems.

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A Appendix

A.1 Hyperparameter Settings

We list the key hyperparameters used for training both AttentionXML and APLC-XLNet models.

A.2 Explainability Examples via LIME

Below are examples of token importance explanations from LIME on the EUR-Lex dataset:

- **Document Excerpt:** "The directive on water quality must be enforced by all EU member states..."
- **Top contributing tokens:** "directive", "water", "EU", "enforced"
- **Predicted Labels:** Environmental Policy, Water Regulation

Table 10: Hyperparameters used for AttentionXML and APLC-XLNet

Parameter	AttentionXML	APLC-XLNet
Learning Rate	0.001	2e-5
Batch Size	256	128
Epochs	20	20
Dropout	0.2	0.1
Optimizer	Adam	AdamW
Embedding Size	300 (FastText)	768 (XLNet)
Label Clusters	3 (PW)	5 (PW)
Adversarial Training	FGSM ($\epsilon = 0.15$)	FGSM ($\epsilon = 0.10$)

A.3 Reproducibility Checklist

- Codebase: GitHub repo (link to be added)
- Preprocessing scripts for EUR-Lex and AmazonCat-13K included
- Trained weights provided in supplementary zip
- Requirements file with library versions included

B Research Methods

Our methodology for evaluating the explainability and robustness of Probabilistic Walk-based Extreme Multi-label Classification (PWXML) models involves several systematic steps, from data processing to final analysis. Below, we outline the complete research pipeline:

B.1 Dataset Preparation

We utilize two benchmark XML datasets:

- **EUR-Lex:** A legal document dataset with multi-label annotations used with AttentionXML.
- **AmazonCat-13K:** A product catalog classification dataset used with APLC-XLNet.

Datasets are preprocessed using `scikit-learn`, `NumPy`, and `pandas` to tokenize text, normalize input formats, and prepare multi-hot encoded label vectors.

B.2 Model Architecture and Training

We implemented and trained the following PWXML models:

- **AttentionXML** with PW clustering and label tree attention, implemented in PyTorch.
- **APLC-XLNet** with Probabilistic Walks and XLNet encoder, using the HuggingFace Transformers library for pre-trained models.

Training was performed on GPU-enabled machines using the PyTorch Lightning framework for efficient model checkpointing and reproducibility.

B.3 Robustness Enhancement and Evaluation

To improve and evaluate adversarial robustness, we applied the following techniques:

- **Adversarial Training** using FGSM (Fast Gradient Sign Method) and PGD (Projected Gradient Descent).

- Simulated attacks using the **Adversarial Robustness Toolbox (ART)** by IBM.
- Evaluations included accuracy under clean vs. adversarial conditions, robustness vs. ϵ plots, and class-specific confusion matrices.

B.4 Explainability and Interpretation

We integrated local explanation methods to understand model decisions:

- **LIME (Local Interpretable Model-agnostic Explanations)** was used to interpret individual predictions and highlight important input tokens influencing label assignment.
- We visualized token-wise importance scores using color-based heatmaps.

All explainability modules were developed using the LIME Python library, along with custom visualizations in `matplotlib` and `seaborn`.

B.5 Performance Metrics

To evaluate effectiveness and fairness, we employed the following multi-label classification metrics:

- **Precision@1**, **Precision@5**
- **Micro F1** and **Macro F1**
- **nDCG@k** (Normalized Discounted Cumulative Gain)

Additionally, for robustness evaluation, we used:

- Accuracy vs. ϵ curves for different adversarial attacks
- Label-wise confusion matrices before and after adversarial training

B.6 Visualization and Insights

- All robustness and explainability results were plotted using `matplotlib` and `seaborn`.
- Experimental logs were tracked using `Weights & Biases (wandb)` for better reproducibility and comparative visualization.

This robust experimental setup enabled us to analyze the trade-offs between performance, robustness, and interpretability in PWXML models across diverse XML tasks.

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